



The potential modulatory role of herbal additives against Cd toxicity in human, animal, and poultry: a review

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Abstract

Cadmium (Cd) is a heavy and toxic metal and easily absorbed by animals and plants; subsequently, it is an environmental risk factor with several toxic effects in humans and animals. The main pathway of human or animal exposure to Cd is through its ingestion by water or food and by particles or fume inhalation during industrial processes. With continuous exposure to small levels of cadmium, it is being deposited in different tissues day after day, causing toxic effects on the liver, kidney, and testes. Long-term exposure to this toxic metal resulted in inflammatory infiltration, necrosis of hepatocytes, degenerative changes in testis tissues, reduction in spermatocytes, degeneration in renal tubules, and hypertrophy of renal epithelium. Therefore, we need an effective treatment to overcome cadmium poisoning. Thus, in the current review, we try to provide compiled reports and summarize information about the toxicological effects of Cd in human, animals, and poultry. This review also provides updated information about the protective actions of herbs and herbal extracts and their role as an effective strategy in reducing or preventing serious health problems and tissue damage in response to Cd toxicity.

Keywords Cadmium toxicity · Heavy metals · Animals · Poultry · Herbs · Therapeutic action

Introduction

The accumulation of excessive quantities of heavy metals induces many intoxication symptoms. Heavy metals including cadmium (Cd) are a serious threat to human and animal health (Abou-Kassem et al. 2016). Humans, animal, and poultry are generally exposed to Cd via food, drinking water, and air. The

contamination of animal with Cd leads to significant economic losses, and therefore, there is growing concern about the environmental pollutants emerging in animal production systems (Galadima et al. 2011; Li et al. 2018). Cd produces various dysfunctions in the body, biochemically and physiologically. Cd accumulation in the body leads to organ dysfunction that manifests as renal proximal tubular cell injury, mild necrosis, inflammation, and hepatic endothelial cells and also causes damage to embryos and the reproductive organs (Thompson and Bannigan 2008). Cd prompts deleterious changes in some functions in the biological systems of body. The Cd intake caused utilization of protein binding sulfhydryl groups, glutathione, and subsequently augments the levels of free radicals (Valko et al. 2005). The main action in the Cd poisoning is that it induces and upsurges the oxygen free radical production in the cells causing oxidative lesions in various tissues, which are indicator for Cd toxicity. Thus, Cd toxicity leads to changes in the antioxidant defense system by altering the levels of lipid peroxidation (Mallya et al. 2017).

Nowadays, Cd toxicity is a common problem fronting the world in all types of living organism such as birds, animals, and mammals including humans, resulting in degradation of the environment and health. There are many articles about Cd toxicity and all recognized this toxicity on all creatures.

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Recently, herbs and herbal extracts are one of the pillars of alternative and complementary medicines (Graham et al. 2005; Alagawany et al. 2017; Dhama et al. 2018), which have exhibited numerous properties against many toxic materials (Bamidele et al. 2012; Yadav et al. 2017; Ewere et al. 2018; Waman et al. 2018). Thus, the approaches of treatment and prevention of Cd toxicity has been focused on the use of herb and its extracts (Abedi et al. 2016). Herbs are able to bind to the heavy metal and reduce its absorption in the intestine, support to release the body's excretion mechanisms (in cases of excessive poisoning), and facilitate its excretion from the body. Also, herbs contain many compounds acting as antioxidants to benefit in amelioration of Cd toxicity in human, animal, and poultry. Therefore, the current review deals with up-to-date information about the toxicological effects of Cd on human, animals, and poultry health, and the potential modulatory role of herbal plants and their extracts against these effects.

Chemical properties and some industrial sources of cadmium

Cadmium is a chemical element with the atomic number 48 and symbol Cd. This metal is chemically in group 12. Cd was discovered in Germany, 1817, by Hermann and Stromeyer as an impurity in zinc carbonate. Figure 1 shows some information and properties of Cd. Cd is present in all soil, air, and aquatic systems. The major natural Cd sources are volcanoes, sea spray, airborne soil particles, forest fires, and biogenic material. The average concentration of Cd in the Earth's crust is about 0.1–0.5 ppm. In non-industrialized zones, Cd

concentration in the air rarely exceeds $0.0025 \mu\text{g}/\text{m}^3$, in soil is about 0.5–1 mg/kg, and in seawater is less than $1 \mu\text{g}/\text{kg}$ (Bersényi 2003). Cd is not usually present as a pure metal in the environment, but most often as complex sulfides, oxides, and carbonates in lead, copper, and zinc ores.

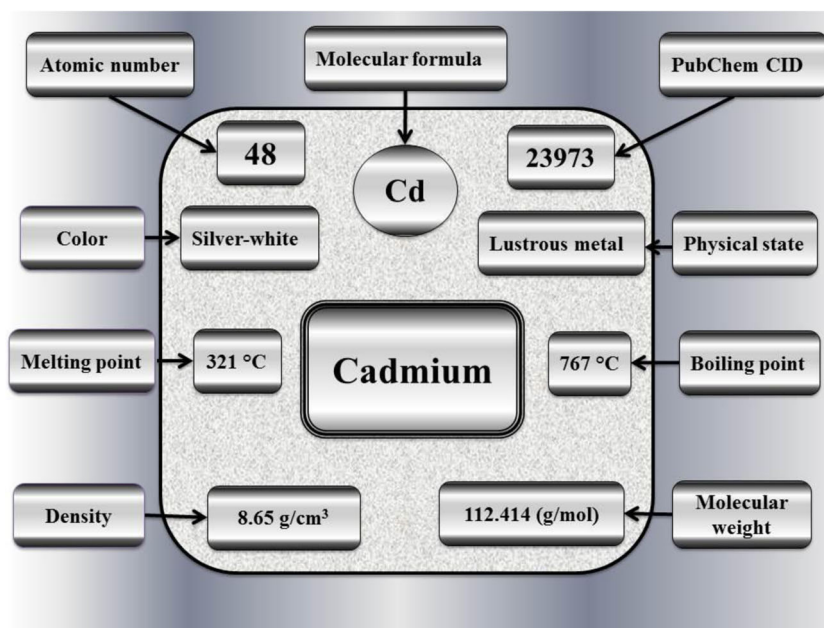
Signs of cadmium exposure

The exposure to any toxic metal has become a recognized source of disease worldwide. Cd, as a heavy metal, is present in the environment and causes some hazard to human and animal. The exposure to Cd has some signs where there are morphological and pathological signs.

Morphological signs

There are some apparent symptoms after cadmium poisoning such as cough, headache, muscle ache, weakness, dizziness, chills, fever, chest pain, irritation and dryness of the throat and nose, pain in the back and joints, osteoporosis, gout, testes tumors, and loss of sense of smell. At very low exposure concentration, Cd is toxic and has chronic and acute effects on the environment and then on human and animal health. After chronic exposure to higher levels of Cd, adverse effects were seen on animals, including decreased feed intake and growth rate (loss of weight) (Bersényi 2003). Additionally, body weight was decreased by increasing Cd level in egg-laying chicken diets (Haziri et al. 2016). Injection of cadmium chloride on the seventh day of incubation in chicken egg albumin ($1.5 \mu\text{g}/\text{egg}$) led to reduction of body weight and prompted significant developmental abnormalities (crown

Fig. 1 Some information and properties of cadmium



rump length, crossed beak, and crooked toes) (Andleeb et al. 2018).

Pathological signs

Several investigations studied the detrimental pathological effect of cadmium on different body organs. Generally, they concluded that cadmium is one of the most toxic and carcinogenic heavy metals that can be distributed mainly to the liver, kidney, and testes, causing severe acute or chronic tissue injury.

Pathological signs in the liver

Brzóska et al. (2003) studied the effect of Cd administration (50 mg) for 12 weeks in rats. Rats given Cd showed ballooning degeneration with high nuclear chromatin density and compact structure of hepatocyte nucleus in mid and periportal zone. However, centrilobular zone showed mononuclear cell infiltrations and single cells necrosis. Koyu et al. (2006) investigated the effect of daily consumption of 15 ppm of Cd in rats for 30 days. Histopathological examination revealed hepatocytes vacuolization and cytoplasmic granules, and sinusoidal and portal dilatation with hyperchromatic nucleus was reported.

Another study was carried out by Amin et al. (2006) who concluded that administration of Cd (3.5 mg/kg BW) to rats led to pathological lesions in the form of mild lobular necrosis and degeneration, nuclear pyknosis and karyolysis, inflammatory infiltration, and congestion in sinusoids and central veins. Moreover, Abdel Moneim and Ghafeer (2007) illustrated that rats injected intraperitoneally with cadmium chloride (0.5 mg/kg) for 4 weeks showed different grades of vascular congestion and dilatation, cellular degeneration, and necrosis of hepatocytes.

Also, Ersan et al. (2008) showed that daily intraperitoneal administration of Cd (0.1 and 0.2 ml) for 15 consecutive days to mice induced dose-dependent necrosis and apoptosis in liver tissue. Similar results were obtained by Gathwan et al. (2012) when Cd was orally administrated to mice at 1–10 mg/kg BW dose levels. Regarding acute Cd toxicity, single IV injection of Cd (4 mg/kg BW) induced pathological alterations in the form of hepatocyte degeneration and centrilobular coagulative necrosis with peripheral hemorrhage surrounding the portal areas (Lee et al. 2009).

Further chronic experiment was conducted by Mahran et al. (2011) who observed that intraperitoneal injection of 0.16 mg CdCl₂/kg BW for 8 weeks resulted in different degrees of hepatocytes degeneration; the cells were enlarged with light foamy cytoplasm. Sinusoidal dilatation, mononuclear infiltration, and Kupffer cell proliferation were also evident. Moreover, few areas of necrosis were observed; cells appeared with pyknotic nucleus and condensed

chromatin. Bamidele et al. (2012) noticed that liver tissues of rats treated subcutaneously with Cd (2.5 mg/kg BW) suffered vacuolar and fatty degeneration with diffused hepatocellular necrosis. Similar hepatocellular picture was obtained by Srinivasan and Ramprasath (2012). Tarasub et al. (2012) treated rats orally with Cd (5 mg/kg) for 27 days. Histopathological examination showed various grades of degeneration, vacuolization, necrosis, and infiltration of inflammatory cell.

Prabu et al. (2012a, b) concluded that rats treated orally with Cd (5 mg/kg BW) of cadmium for 4 weeks showed hepatocellular degeneration including vacuolar degeneration and microvascular steatosis. Besides, hepatocellular necrosis and peri-portal inflammatory infiltration were reported. El-refaiyac and Eissa (2013) studied the pathological effect of Cd intoxication (3 mg Cd/kg BW) on rats' liver. The liver showed severe necrotic and degenerative changes including fatty infiltration, proliferation of bile ductules and Kupffer cells, and focal areas of necrosis. Moreover, Haoem et al. (2013) investigated the effect of cadmium chloride (150 mg/l) addition to drinking water in male rats. Histopathological analysis revealed severe degenerative alteration in liver tissue such as hepatocellular vacuolization, vascular congestion, and polymorphic nuclei. In the same context, Ojo et al. (2014a, b) concluded that treatment with cadmium chloride (4 mg/kg BW) led to severe portal congestion and diffuse cytoplasmic vacuolization.

In 2016, Another investigation was designed by Dardouri et al., and they reported that watery supplementation of cadmium chloride (100 mg/l) in drinking water of rats resulted in marked hepatic injury including vascular dilatation and congestion of central vein and sinusoids, hepatocellular degeneration and vacuolization, focal areas of coagulative necrosis, infiltration of mononuclear inflammatory cell, and proliferation of bile ductules and Kupffer cells. Recently, Alkushi et al. (2018) concluded that subcutaneous injection of 2.5 and 5 mg CdCl₂/kg in rats resulted in congestion and dilatation of central and portal veins, hydropic and granular degeneration, and portal infiltration of mononuclear inflammatory cells. Hepatocyte nucleus appeared pyknotic, karyorrhectic, or completely disappeared. Additionally, different grades of necrosis were observed including single cell necrosis, focal area of coagulative necrosis accompanied with marked liver fibrosis.

Pathological signs in the kidney

Nimotot et al. (1993) injected rats subcutaneously with Cd (0.6 mg kg/day) for 8 weeks. Histopathological examination after 4 weeks revealed weak staining of some proximal tubules indicating mitochondrial depletion of the epithelial lining. Also, several condensed cells were seen in the affected tubules, where it desquamated into the tubular lumen or

pushed out toward the adjacent cells; these cells appeared to have shrunk and had dark eosinophilic cytoplasm and condensed nucleus, suggesting apoptosis. After 8 weeks, regenerated epithelium occupies most of the proximal tubules and the number of apoptotic bodies was slightly increased; however, inflammatory infiltrations, tubular necrosis, glomerular injury, and interstitial fibrosis were not reported.

Brzówska et al. (2003) studied the effect of Cd administration (50 mg) for 12 weeks on rat's kidney. Rats given Cd showed degeneration in renal tubules, hypertrophy of renal epithelium, and glomerular dilatation. Obianime and Roberts (2009) concluded that administration of Cd in the dose range (0–40 mg/kg) caused circulatory disturbance in form of severe local hemorrhage of the renal tissues.

Metwally and Hashem (2009) examined the effect of administration of CdCl₂ (1.5 mg/100 g BW) in rats. Analysis of kidney tissues revealed circulatory disturbances including congestion and focal hemorrhage; besides, focal aggregations of mononuclear cells within the interstitium, hyaline droplet, granular degeneration, and focal necrosis were reported. Also, tubular dilatation, intraluminal cellular casts and eosinophilic material, swollen glomeruli, and mesangial cell proliferation were evident. Siddiqui (2010) concluded that regular supplementation of rat's drinking water with CdCl₂ at a dose of 0.6 mg/kg BW/day for 30 days resulted in marked injury of the renal cortex, proximal and distal tubules, and cellular and brush border membrane; however, glomeruli were the least affected part.

Mahran et al. (2011) studied the impact of IP injection of CdCl₂ (0.16 mg/kg BW) for 8 weeks on rat's kidney. Microscopic examination of renal tissues revealed different grades of tubular damages in the form of hypertrophy and degeneration of tubular epithelium and interstitial infiltration of mononuclear cells. Also, mild epithelial desquamation and enlargement of glomeruli with subsequent narrowing of capsular space were noticed.

Another study was carried out by El-refaiyac and Eissa (2013); microscopic examination of renal tissues from rats intoxicated with Cd (3 mg Cd/kg BW) revealed glomerular shrinkage, degeneration and necrosis of renal epithelium, vascular dilatation, and congestion with focal areas of hemorrhage. Dardouri et al. (2016) studied the pathological impact of cadmium chloride (100 mg/l) watery supplementation in rats; renal tissue from Cd-intoxicated rats showed glomerular shrinkage with subsequent dilatation of capsular space with necrotic and degenerative changes of tubular epithelium.

Mice injected intraperitoneally with cadmium chloride (2 mg/kg/day) for 14 days revealed glomerular lesions including glomeruli atrophy with subsequent dilatation in Bowman's space and also mild interstitial edema and proximal epithelial injury in hematoxylin and eosin-stained sections. However, renal sections stained with periodic acid-Schiff stain (PAS)

showed attenuation in brush border membrane with partial or complete absence of tubular lining epithelium. Moreover, SR-stained section expressed less evident stain around tubules and/or renal glomeruli (Micali et al. 2018).

Pathological signs in the testes

Obianime and Roberts (2009) studied the effect of rat's intoxication with cadmium at dose range (0–40 mg/kg). Light microscope analysis of testicular tissues revealed dose- and time-dependent injury represented by damage of seminiferous tubules and germ cells with subsequent reduction in spermatocytes, interstitial edema and congestion, and scattered necrotic foci.

Another toxicological study was conducted by Metwally and Hashem (2009); they demonstrated that rats given CdCl₂ (1.5 mg/100 g BW) showed marked testicular lesions characterized by degenerative and necrotic changes of some germinal epithelium, with exfoliation of spermatocytes into tubular lumen. Besides, interstitial congestion and edema with multifocal hemorrhagic foci were evident. Also, cauda epididymis had decreased numbers of spermatozoa, with homogenous eosinophilic necrotic debris within epididymal ductule lumen.

Kamel et al. (2011) examined the impact of various concentration of Cd (5 mg/L (low dose) and 50 mg/L (high dose) in drinking water of rats. Microscopically, testes from rats received Cd in low dose showed degenerative alterations vacuolization and desquamation of germinal epithelium accompanied by interstitial edema. Moreover, intraluminal aggregations of multinucleated giant cells were noticed. Moreover, testicular tissues from rats that received Cd in high dose showed similar pathological lesions including germinal vacuolization and necrosis, with total absence of spermatozoa, interstitial hemorrhage and edema, and Leydig cell necrosis.

In the same line, rats treated with 3 mg Cd/kg BW showed marked testicular lesions represented by interstitial edema and congestion, degeneration of germinal epithelium, and with dramatically reductions in the number of sperms in the lumen of seminiferous tubules (El-Refaiyac and Eissa 2013). Fouad and Jresat (2014) illustrated that IP injection of cadmium chloride (2 mg/kg) in rats resulted in massive necrosis and vacuolization of germinal epithelial lining with subsequent impairing of spermatogenesis; also, different grades of interstitial hemorrhage and edema were reported.

Another subchronic study was designed by Adamkovicova et al. (2017); they concluded that intoxication of rats with Cd (30 mg/L) for 90 consecutive days led to lumen contraction of seminiferous tubules, with degeneration of germinal epithelium represented by vacuolization and desquamation. However, intertubular interstitium appeared unaffected except for mild dilatation and congestion of interstitial vasculature. Jahan et al. (2014) concluded that subcutaneous administration of CdCl₂ (0.2 mg/kg) led to dilatation of tubular lumen,

sloughing of germinal epithelium with subsequent reduction in sperm numbers, and degenerative changes within interstitial tissues.

In further acute toxicological investigation, Kheradmand et al. (2015) reported that single IP injection of cadmium chloride (2 mg/kg) resulted in massive destruction of germinal epithelium with resultant reduction in spermatogenesis; only a layer of spermatogonia was detected in seminiferous tubules; also, interstitial hemorrhage was reported. Alkhedaide et al. (2016) found that administration of CdCl₂ (5 mg/kg BW) induces interstitial congestion and edema in testicular tissues of rats. Moreover, degeneration and shedding of germinal epithelium and corrugation of seminiferous tubular epithelial lining were evident.

In another work, testes from rats intoxicated with CdCl₂ (2.0 mg/kg) showed marked degeneration, vacuolization, and sloughing of germinal epithelium (Liu et al. 2016). Ansa et al. (2017) examined the effect of Cd intoxication (3 mg CdCl₂/kg/day, for 7 days) on rabbit testes, and they reported that rabbits intoxicated with Cd developed marked testicular alterations represented by dilatation of seminiferous tubule lumen, vacuolization, and sloughing of germinal epithelium with subsequent impairment in spermatogenesis process; also, interstitial hemorrhage was reported. Similar lesions were obtained by Babaknejad et al. (2018) after intoxication of male rats with 1 mg Cd/kg BW.

Detoxification activities of some herbs and plants

Major findings of previous investigations concerning using of phytomedicinal plants against cadmium-induced toxicities are summarized in Table 1 and Fig. 2.

Black cumin (*Nigella sativa*)

Nigella sativa is an annually flowering herb with various clinical benefits. Several investigations concluded the antioxidant ameliorative effects of *Nigella sativa* against Cd-induced oxidative stress pulmonary injury (El-Ebiary et al. 2016), and adverse effects on fertility and reproduction function of male albino rats (Hassanin and Safwat 2014). Administration of *Nigella sativa* oil with CdCl₂ exhibited a significant enhancement in the tested histopathological and biochemical indices of rats (Mohammed et al. 2014). *Nigella sativa* acts as protective impact against Cd toxicity (Kanter et al. 2005). This antioxidant ability may be attributed to its increased contents of the active ingredient thymoquinone. Fouad and Jresat (2014) illustrated that thymoquinone may signify a possible therapeutic choice to safeguard against the toxic effect of Cd exposure on the testis.

Winter cherry (*Withania somnifera*)

Withania somnifera is a common medicinal herb that has been intensively used in many researches concerning attenuation and elimination of cadmium toxicity in broiler chickens. Oral administration of winter cherry root prevented Cd-induced peroxidation of chicken tissues (Bharavi et al. 2010). It was reported that *Withania somnifera* was able to ameliorate the impact of cadmium-induced oxidative stress and hepatorenal injury in broilers (Bharavi et al. 2010; Bharavi et al. 2011; Yadav et al. 2017), which may be due to its great contents of alkaloids, steroidal lactones, tropine, withanolides, and withaferin A.

Korean red ginseng extract (*Panax ginseng*)

Panax ginseng, a common species of “Ginseng” plant family, is widely and effectively used to improve immunity and attenuate the side effects of different types of hazardous treatments. Also, it was successfully used against cadmium-induced tissue oxidation and injury. Ginsenosides in *Panax ginseng* play a pivotal role in its antioxidant and antiapoptotic properties. Previous investigations reported antioxidant, antiapoptotic, and hepatoprotective ability of *Panax ginseng* against cadmium toxicity in rats (Karadeniz et al. 2009; Park et al. 2013), mice (Shukla and Kumar 2009), and laying hens (Park et al. 2013).

Black and green tea (*Camellia sinensis*)

Black and green tea are two species that come from *Camellia sinensis* leaves; they both have beneficial medicinal activities. Winiarska-Mieczan (2015) examined the impact of using black and green tea in rats intoxicated with cadmium; they concluded the role of green and black tea in eliminating and reducing metal accumulation in tissues. *Camellia sinensis* have a protective impact on the blood profile of rats against the Cd toxicity (Singh et al. 2013). Also, green tea possesses useful effect in ameliorating Cd toxicity in ovarian tissue (Soares et al. 2013). Consumption of green tea extract by rats exposed to Cd poisoning can effectively lessen the cell injury in the liver and kidney (Hussain and AL-Tae 2014). The protective role played by black and green tea may be attributed to their antioxidant activity indicated by elevated SOD, CAT, and GPx in all organs (Winiarska-Mieczan 2015).

Cinnamon (*Cinnamomum zeylanicum*)

Recently, cinnamon has been emerging as a new herbal supplement with several beneficial health impacts including anti-inflammatory, antimicrobial, and cardioprotective properties. Concerning its effect against cadmium-induced oxidative damage and tissue injury, Ibrahim and Abo-Kora (2018)

Table 1 Major findings of previous investigations concerning the use of phytochemical plants against cadmium-induced toxicities

Herb	Common name	Extract/active ingredient	Study type	Form of Cd	Mechanisms of action	References
<i>Nigella sativa</i>	Black cummin	Thymoquinone	In vivo	Cadmium chloride (CdCl ₂) 2 mg/kg BW	• Exert protective role against cadmium-induced oxidative stress in rats	Coskun and Gurel (2005)
<i>Withania somnifera</i>	Ashwagandha, Indian ginseng, winter cherry	Alkaloids, steroidal lactones, tropine, cuscohygrine, withanolides, withaferin A	Adult rats	Cadmium chloride (CdCl ₂) 5 mg/kg BW	• Ameliorates cadmium-induced elevation male reproductive hormones and inflammatory markers (TNF-α and IL-1β) • Antioxidant protective effect against Cd-induced lung injury in rats • Significant amelioration of cadmium-induced oxidative damage and improvement in body weights and liver and kidney functions	Hasanin and Safwat (2014)
			In vivo	Cadmium chloride (CdCl ₂) 2 mg/kg BW Cadmium chloride monohydrate (100 ppm)	• Antioxidant protective efficacy against Cd-induced liver and kidney damages and oxidative stress	El-Ebiary et al. (2016)
			Broiler chicks	Cadmium chloride (100 ppm)	• Possesses antioxidant protective efficacy against Cd-induced liver and kidney damages and oxidative stress	Bharavi et al. (2011)
<i>Ocimum sanctum</i>	Holy basil, tulasi	Eugenol	In vivo	Cadmium chloride monohydrate (100 and 200 ppm)	• Significant amelioration of serum biochemical parameter (protein, albumin, globulin, ALT, AST, uric acid, and creatinine) in cadmium-intoxicated chicks	Yadav et al. (2017)
<i>Asperagus recemosus</i>	Satavar, shatavari, or shatamull	Saponins, isoflavones, asparagamine, racemosol, polysaccharides, mucilage	Broiler chicks	Cadmium chloride (100 ppm)	• Reduction in hepatic and renal Cd accumulation • Reduction in Cd-induced peroxidative tissues damage	Bharavi et al. (2011)
<i>Andrographis paniculata</i>	Green chiretta, Kalmegh	Andrographolide			• Antioxidants scavenger effect against oxygen free radicals • Chelating and elimination activity against Cd	
<i>Asphaltum panjabinum</i>	Shilajith	Asphaltum punjabinum				
<i>Gymnema sylvestre</i>	Gumar, merasingi, meshashringi, gumarbooti, peiploca of the woods, rams' hom, small	Gymnemic acids, gymnemasaponins, polypeptide, gumarin				
<i>Spirulina platensis</i>	Indian ipecac	Cyclic peptides, alkaloids and lipopolysaccharides	In vivo female Wistar albino rats	Cadmium chloride (CdCl ₂)	• Ameliorates cadmium-induced hepatotoxicity due to its antioxidant properties.	Karadeniz et al. (2009)
			In vivo Swiss albino mice	Cadmium chloride (CdCl ₂) 1.0 mg/kg WB	• Ameliorates Cd-induced tissue peroxidation	Shukla and Kumar (2009)
			In vivo broiler chicks	Cadmium chloride (100 ppm)	• Possesses antioxidant protective efficacy against Cd-induced liver and kidney damages and oxidative stress	Bharavi et al. (2011)
			In vivo rats			Gaurav et al. (2011)

Table 1 (continued)

Herb	Common name	Extract/active ingredient	Study type	Form of Cd	Mechanisms of action	References
<i>Panax ginseng</i>	Korean red ginseng extract	Ginsenosides	In vivo female Wistar albino rats	Cadmium chloride (CdCl ₂) 2 mg/kg WB	- Hepatoprotective and renoprotective effect through the antioxidant capabilities - Ameliorates cadmium-induced hepatotoxicity	Karadeniz et al. (2009)
			In vivo Swiss albino mice	Cadmium chloride (CdCl ₂) 1.0 mg/kg BW	Ameliorates Cd-induced tissue peroxidation due to its antioxidant properties	Shukla and Kumar (2009)
			In vivo broiler and laying chicks	Cadmium chloride (100 ppm)	Attenuates Cd-induced hepatic toxicity, through antioxidant and antiapoptotic properties	Park et al. (2013)
			In vivo rats	Cadmium chloride (CdCl ₂) 2 mg/kg BW		
<i>Camellia sinensis</i>	Black, red and green tea	Caffeine	In vivo Rats	Cd (7 mg/kg BW)	- Significant reduction in tissue accumulation of Cd - Significant elevation in the activity of SOD, CAT, and GPx in all organs of rats	Winiarska-Mieczan (2015)
	Cinnamon	Cinnamaldehyde, cinnamyl acetate, eugenol, tannins	In vivo Albino rats	Cadmium chloride 20 mg/kg/day	Ameliorates Cd-induced testicular damages	Ibrahim and Abo-Kora (2018)
<i>Corchorus olitorius</i>	Edible jute	Phenolic compounds and flavonoids	- In vitro isolated mice hepatocytes - In vivo rats	- CdCl₂ (30 µM) - Cd Cl₂ (4 mg/kg BW)	- Improvement in hematological, biochemical, and histologic antioxidant parameters, and histologic structure of tissues - Ameliorates DNA damage and cd accumulation inside cells	Devanajee et al. (2013)
<i>Cynara scolymus</i>	Artichoke	Cynarin, luteolin, cynardoside	In vivo	Cadmium chloride (1 mg/100 g)	Ameliorates Cd-induced testicular damage and decrease level of NO production	Gurel et al. (2007)
			Male albino Wistar rats	CdCl ₂ (100 mg/L drinking water)	Ameliorates Cd-induced hepatorenal oxidative damage and immunosuppression	El-Boshy et al. (2017)
<i>Emblica officinalis</i>	Amla, Indian gooseberry	Lannins (gallic acid, ellagic acid)	In vivo Rats	CdCl ₂ (3 mg/kg BW)	Exerts cytoprotective effect against Cd-induced acute toxicity which may be attributed to its antioxidant contents.	Khandelwal et al. (2002)
			In vivo One-day-old male broiler chicks	Cd (100 ppm in feed)	Had antioxidant effect against Cd-induced toxicity	Swapna et al. (2010)
			In vivo Swiss albino mice	Cadmium chloride (20 ppm)	Ameliorates radiation and Cd-induced neurotoxicity	Purohit et al. (2011)
<i>Allium sativum</i> L.	Garlic	Diallyl sulfide, diallyl disulfide, and diallyl trisulfide	In vivo	Cadmium chloride monohydrate (0.5 mg/kg/4 ml, IP once a day)	Ameliorates Cd-induced increase in tissue zinc and copper due to depletion of metals toxic effect.	Tandon et al. (2001)
			Adult albino rats	CdCl ₂ (1.5 mg/100 g BW)	Enhancement of antioxidant properties and ability to chelating metals	Metwally and Hashem (2009)
					Ameliorates Cd-induced oxidative stress	Eteng et al. (2012)

Table 1 (continued)

Herb	Common name	Extract/active ingredient	Study type	Form of Cd	Mechanisms of action	References
<i>Hibiscus sabdariffa</i>	Malvaceae	Anthocyanin	In vitro Human embryonic kidney cells (HEK 293)	CdCl ₂ single acute dose (3 mg/kg BW) Cadmium chloride (200 ppm in drinking water)	• Reduction in Cd accumulation by reduces its absorption and improves its excretion	Nwokocho et al. (2014)
			In vivo Juvenile sea bass specimens	Cadmium chloride (CdCl ₂)	• Exerts anti-Cd properties, potent scavenging radical and chelating activity	Boonpeng et al. (2014)
			In vivo Male Wistar albino rats	CdCl ₂ (500 µg/kg) Cadmium chloride (CdCl ₂) Cadmium chloride (S/C 0.002 g)	• Ameliorates Cd-induced immunosuppressive effects in liver and head kidney • Ameliorates Cd-induced hepatotoxic effect due to its antioxidant properties • Ameliorates Cd-induced oxidative damages due to its antioxidant properties	Mosbaha et al. (2017) Asagba et al. (2007) Omonkhua et al. (2009)
<i>Irvingia gabonensis</i>	wild mango, African mango, Bush mango		In vitro (Human u937 cell)	Cadmium chloride (0.1 µ mol/L)	• Ameliorates Cd-induced immunosuppressive effect due to its antioxidant properties	Okoko and Ere (2012)
			In vivo Rats	Cadmium chloride (4 mg/kg/day, IP)	• Possesses renoprotective effect against Cd-induced nephrotoxicity	Ojo et al. (2014)
			In vivo Male Wistar rats In vivo Female Wistar rats	Cadmium chloride (CdCl ₂) Cadmium chloride (10 mg/kg BW)	• Shows hepatoprotective and antioxidant activity against Cd-induced toxicity • Exerts hepatonephroprotective effect against Cd-induced toxicity • Possesses hypolipidemic effect against Cd-induced toxicity	Ewere et al. (2016) Ewere et al. (2018)
<i>Momordica charantia</i> <i>Muraya koenigii</i>	Bitter melon, bitter apple; bitter gourd Curry leaf	Momordin and charantin Carbazole alkaloids	In vivo Male Wistar rats In vivo Rats	Cadmium chloride (2.5 mg/kg BW) Cadmium chloride (low dose)	• Possesses enhancement effect on LH, ALP, LDH, and G-6-PDH levels • Possesses cardioprotective effect against cadmium-induced oxidative stress due to antioxidant properties	Bamidele et al. (2016) Mitra et al. (2012)
			In vivo Male albino rats	Cadmium chloride (CdCl ₂ S/C)	• Ameliorates Cd-induced oxidative damages due to its antioxidant properties	Mitra et al. (2013a, b)
			In vivo Albino rats	CdCl ₂ (6.0 mg/kg BW)	• Possesses antioxidant effect against Cd indicated by significant elevation in LPO levels and significant reduction in levels of GSH, GPx, SOD, CAT, and ascorbate	Ramesh and Satakopan (2010)
<i>Ocimum sanctum</i>	Tulsi leaf Indian holy basil-tulsi	Cirsilineol, cirsimartin, isothymusin, apigenin, and rosmarinic acid	In vivo Male broiler chicks	Cadmium chloride (100 ppm)	• Significant improvement in body weight • Prevention of Cd bioaccumulation of and tissue oxidative damage	Bharavi et al. (2011)
			In vivo	Cadmium chloride (CdCl ₂ S/C)	• Possesses antioxidant effect against Cd-induced liver toxicity	Mitra et al. (2013a, b)
			Male albino rats		• Possesses antioxidant effect against Cd-induced cardiac toxicity	Mitra et al. (2014)

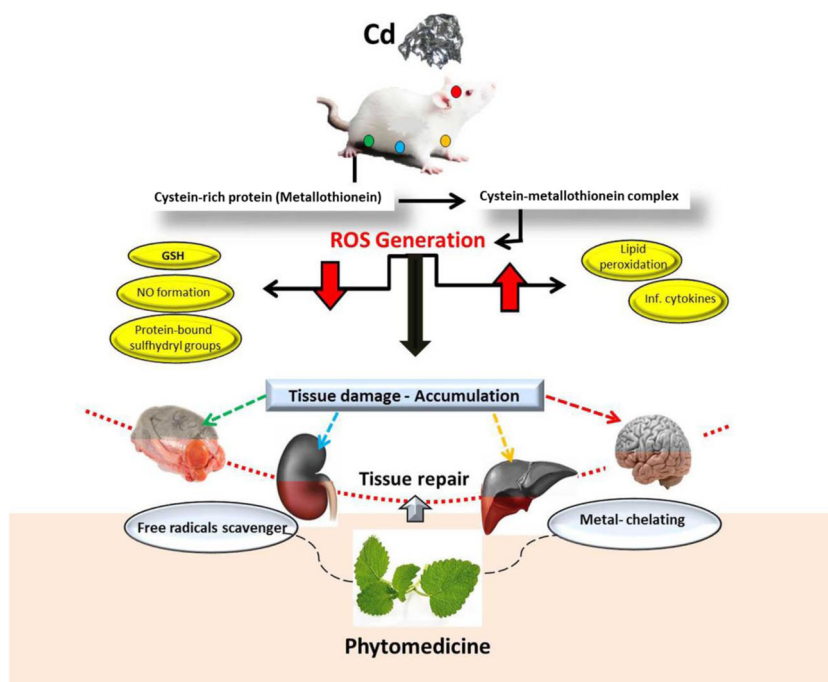
Table 1 (continued)

Herb	Common name	Extract/active ingredient	Study type	Form of Cd	Mechanisms of action	References
<i>Origanum majorana</i>	Majoram	5,6,3'-Trihydroxy-7,8,4'-trimethoxyflavone, hesperetin, hydroquinone, arbutin, and rosmarinic acid	In vivo Albino rats	CdCl ₂ (5 mg/kg BW)	• Showed protective effect on Cd-induced hepatotoxicity and nephrotoxicity via its content of phytochemicals	Shati (2011)
<i>Physalis peruviana</i>	Cape gooseberry	Physalins A, B, D, and F and glycosides	In vivo Adult male Wistar rats	Cadmium chloride (6.5 mg/kg B.W.)	• Possesses ameliorating effect against Cd-induced oxidative neurotoxicity. • Possesses ameliorating effect against Cd-induced testicular damages.	Abdel Moneim et al. (2014) Othman et al. (2014)
<i>Piper betle</i>	Betel	Tannins, chavicol, phenyl, propane, sesquiterpene	In vivo Rats	Cd (5 mg/kg BW)	• Showed ameliorative effects against Cd-induced hepatic oxidative stress.	Prabu et al. (2012a, b)
<i>Psidium guajava</i>	Common guava	Pentacyclic triterpenoid guajanoic acid, s beta-sitosterol, uvaol, oleanolic acid, and ursolic acid	In vivo male albino rats	Cadmium chloride (CdCl ₂)	• Ameliorates Cd-induced hepatotoxicity via its antioxidant constituents	Oa et al. (2013)
<i>Rosmarinus officinalis</i>	Rosemary	Rosmarinic acid, ursolic acid, and oleanolic acid	In vivo male rats	Cadmium chloride (15 mg/kg BW) Cadmium chloride (230 mg/kg BW) Cadmium chloride (5–10 ppm)	• Ameliorates Cd-induced nephron-hepatotoxicity and oxidative injury • Ameliorates Cd-induced oxidative stress and tissue injury • Ameliorates Cd-induced oxidative stress via its antioxidant constituents	Virk et al. (2013) Sakr et al. (2015) Al-Anazi et al. (2015)
<i>Satureja hortensis</i>	Summer savory or garden savory	Carvacrol, thymol, phenols, and flavonoids	In vivo male rats	Cadmium chloride (CdCl ₂)	• It can be concluded that savory extract significantly restore serum amylase and lipase in cadmium-exposed rats and could protect pancreatic tissue in such rats	Khorasani et al. (2013)
<i>Silybum marianum</i>	Milk thistle, cardus marianus, silymarin	Silibinin, silidianin, and silichristin	In vitro human lung carcinoma cells (A549)	Cadmium chloride (3 mg kg BW)	• Ameliorates Cd-induced oxidative stress	Khorasani and Adibi (2012)
<i>Terminalia chebula</i>	<i>Myrobalan</i>	Ayurvedic formulation	In vivo rats	Cadmium chloride (3 mg kg BW)	• Possesses ameliorative efficacy on the viability and DNA damage in A549 cells	Pirbalouti (2015) Webster et al. (2013)
<i>Thunbergia laurifolia</i>	Blue trumpet vine	Proanthocyanidin	In vivo Wistar rats	Cadmium chloride (1 mg kg BW)	• Ameliorates Cd-induced nephrotoxicity and oxidative stress	Thiruchelvi et al. (2012)
<i>Thymus vulgaris</i>	Garden thyme	Thymol	In vivo albino rats	Cadmium chloride (50 PPM in D.W)	• Ameliorates Cd-induced hepatonephrotoxicity and oxidative stress	Ruangyutikarna et al. (2013)
				Cadmium chloride (3.5 mg kg BW)	• Possesses neuroprotective effect against Cd-induced neurotoxicity via its antioxidant properties	Ulwai et al. (2010)
<i>Tribulus terrestris</i>	Goat's head, bindii, bullhead, burra gokharu	Saponins, flavonoids, glycosides, alkaloids, and tannins	• In vitro • In vivo Wistar adult male rats	Cadmium chloride (3 mg kg BW)	• Possesses hepatoprotective effect against Cd-induced hepatotoxicity via its antioxidant capabilities	Elgaml and Hashish (2014)
			In vivo rats	Cadmium chloride (CdCl ₂)	• Ameliorates Cd-induced testicular damages and oxidative injury via antioxidant and metal chelating properties	Rajendar et al. (2011)
						Lakshmi et al. (2012)

Table 1 (continued)

Herb	Common name	Extract/active ingredient	Study type	Form of Cd	Mechanisms of action	References
<i>Zingiber officinale</i>	Ginger	Gingerol, shogaol, citral, pyrogallol	In vivo adult rats	Cadmium chloride (200 ppm)	- Shows potent protective efficacy against Cd-induced oxidative stress - Therapeutic ginger protects against Cd-induced hepatic damages better than prophylactic one	Egwurugwu et al. (2007)
				Cadmium chloride (100 ppm)	Ameliorates Cd-induced hepatotoxicity via decrease bioavailability and metal elimination	Chukwuemeka et al. (2012)
				Cadmium chloride (3 mg kg BW)	Ameliorates Cd-induced oxidative stress	Eteng et al. (2012)
				Cadmium bromide (100 mg)	Possesses protective effect against cadmium bromide induced liver and kidney histopathological alterations	Sarbaz et al. (2013)
				Cadmium chloride (5 mg kg BW)	Ameliorates Cd-induced nephrotoxicity and histologic alterations	Gabrab et al. (2017)
			In vivo male rabbits	Cadmium chloride (200 mg kg BW)	Ameliorates Cd-induced genetic response, and immunohistochemical expression of MKI67 and caspase-3 in the liver and kidney	Baiomy and Mansour (2016)
			In vivo pregnant female Sprague-Dawley rats	Cadmium chloride (8.8 mg kg BW)	Ameliorates Cd-induced toxic effect on fetal and maternal liver and kidney tissues	El-Aziz et al. (2018)

Fig. 2 The possible molecular mechanisms of cadmium-induced oxidative injury and tissue damage



studied the effect of cinnamon against cadmium-intoxicated rats; they conclude beneficial role of cinnamon in the attenuation of Cd-induced testicular damages. In recent study, Muhammad et al. stated that cinnamon bark oil has protective impact against the cadmium-induced toxic effects in the testes of rat. They clarified that administration of cinnamon bark oil protected testicular toxicity induced by cadmium through increasing the germinal epithelium thickness and improvement of spermatogenic cells.

Edible jute (*Corchorus olitorius*)

Corchorus olitorius has been utilized in traditional medicine against several diseases. Several studies reported the nutritional and medicinal role of *Corchorus olitorius* supplements. Phenolic compounds and flavonoids represent a major component of this herb, and this fact explains why this plant is used as an efficient antioxidant. In the same context, Dewanjee et al. (2013) reported improvement in histologic structure and hemato-chemical parameters. Besides, enhancement of antioxidant parameters and attenuation of metal accumulation inside cells were reported. They indicated that the protective action of *Corchorus olitorius* signifies a promising method for the defense against Cd intoxication.

Artichoke (*Cynara scolymus*)

Artichoke leaf extract has been reported previously for its potential effect on cholesterol levels in animals and human. Recently, it begins to be used against several drugs and heavy

metal that induce oxidative stress in animal and human tissues. Cynarine contained within artichoke play the major role in its antioxidant ability. Gurel et al. (2007) and El-Boshy et al. (2017) demonstrated the protective role of artichoke against testicular and hepatorenal oxidative injury induced in rats by Cd intoxication, respectively.

Indian gooseberry, amla (*Emblica officinalis*)

Emblica officinalis is an Indian medicinal herb which is used traditionally and efficiently to enhance general health condition and cognitions. Many researchers have used *Emblica officinalis* to ameliorate the detrimental impact of heavy metal cadmium on body functions; neuroprotective effect of *Emblica officinalis* against Cd was reported in mice by Purohit et al. (2011), while its antioxidant properties and ROS scavenging ability were previously cleared in both rats (Khandelwal et al. 2002) and broiler chicks (Swapna et al. 2010). Using *Emblica officinalis* extract (200 mg/kg body weight) introduced more antioxidant effect for the Cd-induced cellular damage in rats (Virk et al. 2013). Gallic acid and ellagic acid has been found to have the primary role in its antioxidant activities.

Garlic (*Allium sativum* L.)

Garlic is among the most effective medicinal herbs due to its high content of vitamin C, diallyl sulfide, diallyl disulfide, and diallyl trisulfide, which make it a potential candidate in immune enhancing and antioxidant properties. Several research

conclude efficient role of garlic against many hazards such as heavy metals. Prior in vitro (Boonpeng et al. 2014) and in vivo studies concluded the antioxidant, immune enhancing, and metal chelating effect of garlic (Tandon et al. 2001; Metwally and Hashem 2009; Eteng et al. 2012; Nwokocho et al. 2014; Mosbaha et al. 2017). Study of Andleeb et al. (2018) authenticated the strong ameliorative nature of *Allium sativum* against Cd and recommended its use in any method at regular source to resist the harmfulness of Cd in water and food. They stated body weight in the group treated with Cd and garlic was higher than the group treated with Cd only, and this indicates the role of garlic in resisting the effect of cadmium poisoning.

Malvaceae, roselle (*Hibiscus sabdariffa*)

Hibiscus sabdariffa is a common herbal plant which reported to its ability in preventing carbohydrate absorption which resulted in reducing blood pressure. Also, it plays a key role in many toxicological oxidative studies; protective role of *Hibiscus sabdariffa* against cadmium-induced tissue oxidation and immune suppression was evaluated in vivo in rats (Asagba et al. 2007 and Omonkhua et al. 2009) and in vitro (Okoko and Ere 2012) in human u937 cell. The administration of *Hibiscus sabdariffa* extract lessened the toxic effects prompted by Cd (Muhammad and Charles 2008). Also, the study of Orororo et al. (2018) confirmed the antioxidant ability of *Hibiscus sabdariffa* in ameliorating Cd-prompted oxidative stress.

Wild mango, African mango, bush mango (*Irvingia gabonensis*)

Fruits of *Irvingia gabonensis* are similar to mango; it is known for its high content of fiber which have a role in the breakdown and removal of cholesterol from body. Also, it plays a role in toxicological studies, where it is used by many researchers to ameliorate the hepatic and renal oxidative injury induced experimentally in rats by cadmium chloride (Ojo et al. 2014a, b; Ewere et al. 2016). Recently, it possesses hypolipidemic effect against cadmium-induced toxicity in rats (Ewere et al. 2018). *Irvingia gabonensis* has a defensive influence on blood profile against the Cd toxicity (Ojo et al. 2014a, b).

Bitter melon, bitter apple, bitter gourd (*Momordica charantia*)

For long times, *Momordica charantia* is a part of phytomedicinal traditional system. Mainly, it has been used for gastric complaint and treatment of diabetes mellitus. Recently, it is reported for its hepatoprotective effect against various drugs and heavy metals including cadmium-induced hepatotoxicosis (Bamidele et al. 2016). Additionally,

Momordica charantia leaves possess potentials to return disturbed levels of testicular marker enzymes and gonadotrophic hormones detected in Cd-induced gonad damage (Ajilore et al. 2016).

Curry leaf (*Murraya koenigii*)

Murraya koenigii is a traditional commonly used herb which is known widely in India and other countries as a spice. In medicine, it possesses ameliorating characteristic against stomach and intestinal disturbance. Now, it begins to be used in toxicological study as a remedy for many drugs and heavy metals due to its carbazole alkaloid content which is responsible for its antioxidant properties (Mitra et al. 2012, 2013a, b). Feeding the rats on curry leaf extract (100 mg/kg body weight) improved the changes caused by cadmium (Yadav and Saxena 2017). Also, administration of rats with curry leaf extract may be helpful in reducing Cd-induced oxidative injury in the liver (Mitra et al. 2013a, b).

Indian holy basil-tulsi, tulsi leaf (*Ocimum sanctum*)

For thousands of years, *Ocimum sanctum* has been cultivated for health and religious causes. Several investigations were designed to study the ameliorating effect of *Ocimum sanctum* against cadmium-induced oxidative stress and tissue injury. It was concluded that *Ocimum sanctum* possesses antioxidant properties and tissue repair, with prevention of metal bioaccumulation and improvement of oxidative status after intoxication with cadmium (Ramesh and Satakopan 2010; Bharavi et al. 2011; Mitra et al. (2013a, b & 2014). The treatment with *Ocimum basilicum* extract improved histomorphometrical and immunohistochemical alterations induced by Cd, and this could be attributed to its antioxidant ability (Sakr and Nooh 2013).

Marjoram (*Origanum majorana*)

Marjoram is commonly known cooking spice. However, its flower, leaves, and oil extract were used in medicine to treat signs of menopause. Moreover, the antioxidant properties of this plant were reported in many toxicological studies. In rats intoxicated with cadmium, the hepatic and renal toxicity was alleviated by using of marjoram (Shati 2011).

Cape gooseberry (*Physalis peruviana*)

Physalis peruviana is a common commercial plant that is known for its anti-inflammatory properties. Several toxicological studies were designed to investigate the neuroprotective and reproductive enhancement effect of *Physalis peruviana* against cadmium-induced neurotoxicity and reproductive impairment (Othman et al. 2014).

Betel (*Piper betle*)

Piper betle is an important herbal plant that is originally native to Malaysia. Recently, it began to be used as an antioxidant against various oxidative hazards including drugs and heavy metals. The desirable antioxidant effect of *Piper betle* against cadmium-induced hepatic oxidative damages may be attributed to tannins, chavicol, phenyl, propane, and sesquiterpene contained in the plant (Prabu et al. 2012a, b).

Common guava (*Psidium guajava*)

Psidium guajava is a common medicinal plant that is originally native to South America. It was traditionally used in medicine as antioxidant agent where it is able to ameliorate the oxidative hazards. The desirable antioxidant effect of *Psidium guajava* against cadmium-induced hepatic oxidative damages may be attributed to presence of pentacyclic triterpenoid guajanoic acid, β -sitosterol, uvaol, oleanolic acid, and ursolic acid in the plant. Oa et al. (2013) concluded the hepatoprotective and antioxidant effect of *Psidium guajava* against hepato-oxidative injury induced by cadmium.

Rosemary (*Rosmarinus officinalis*)

Strong anti-inflammatory and antiseptic effect of rosemary was previously reported, which explains the presence of flavonoids and phenolic acids in this plant. Moreover, rosmarinic acid extracted from the plant has strong antioxidant properties against various oxidative injuries. Several studies were designed to investigate the in vivo effect of *Rosmarinus officinalis* against hepatic, renal, and oxidative stress induced by cadmium chloride in rats (Virk et al. 2013 and Sakr et al. 2015) and Nile tilapia (Al-Anazi et al. 2015).

Summer savory or garden (*Satureja hortensis*)

Satureja hortensis is traditional popular plant which is originally native to Atlantic Canada. Recently, *Satureja hortensis* has emerged as an efficient antioxidant agent. Many researchers concluded its ability as a scavenger for reactive oxygen species. The ameliorative antioxidant impact of *Satureja hortensis* has been reported against cadmium-induced oxidative injury in rat organs (Khorasgani and Adibi 2012; Khorasgani and Pirbalouti 2015) particularly in pancreatic tissues (Khorasgani et al. 2013).

Cardus marianus, milk thistle (*Silybum marianum*)

Silybum marianum is a popular herbal plant that is rich in silibinin, silidianin, and silichristin with subsequent efficient

antioxidant activity. This antioxidant property has been used extensively to ameliorate the potential in vivo damaging effects of several oxidative agents, in addition to its role in the treatment of cancer and liver disease. In vitro effect of *Silybum marianum* has been studied previously against cadmium-induced DNA damage in human lung carcinoma cells (A549) (Webster et al. 2013).

Myrobalan (*Terminalia chebula*)

Terminalia chebula is among the most effective medicinal herbs due to its excellent content of Ayurvedic formulation which renders it as a potential antioxidant candidate. Several researches conclude the efficient role of *Terminalia chebula* against many hazards such as heavy metals. Thiruchelvi et al. (2012) concluded the renoprotective and antioxidant ability of *Terminalia chebula* against cadmium-induced nephron-oxidative damage.

Blue trumpet vine (*Thunbergia laurifolia*)

Thunbergia laurifolia is an Indian medicinal herb which is used traditionally and efficiently to enhance the general health condition. Many researchers have used *Thunbergia laurifolia* to ameliorate the detrimental effect of oxidative agents on body functions; hepatic and renal protective effect of *Thunbergia laurifolia* against cadmium-induced oxidative stress was reported in rats by Ruangyuttikarna et al. (2013). This antioxidant property may be attributed to the excellent amount of proanthocyanidin contained in such plant.

Garden thyme (*Thymus vulgaris*)

Thymus vulgaris has emerged as one of the most efficient phytomedicinal herbs due to its high content of thymol, which is a potential strong antioxidant agent. Also, this plant possesses anti-inflammatory, antiseptic, and antimicrobial medicinal impacts. Several studies conclude the efficient role of *Thymus vulgaris* against many hazards such as drugs and heavy metals; from them, Ulaiwi et al. (2010) has concluded the protective role of *Thymus vulgaris* against cadmium-induced neurotoxicity in rats. Also, the hepatoprotective ability of this plant against cadmium was suggested by Elgaml and Hashish (2014).

Goat's head, bindii, bullhead, burra gokharu (*Tribulus terrestris*)

Roots and fruits of *Tribulus terrestris* are known for its positive effect on male fertility and urogenital health enhancement; moreover, it can be used to improve the

general status of cardiovascular system. Also, the great content of saponins, flavonoids, glycosides, alkaloids, and tannins in such plant renders it as a potential antioxidant candidate; parallel to this fact, Rajendar et al. (2011) and Lakshmi et al. (2012) concluded the efficient protective role of *Tribulus terrestris* against cadmium-induced testicular and oxidative damages, respectively.

Ginger (*Zingiber officinale*)

For many years, *Zingiber officinale* has been used as a safe remedy for gastric upset. In addition, antioxidant properties of ginger were a key word for many investigations. The antioxidant and the hepatorenal protective effects of ginger against cadmium-induced oxidative injury were previously evaluated in rats (Egwurugwu et al. 2007; Chukwuemeka et al. 2012; Eteng et al. 2012; Sarbaz et al. 2013; Gabrab et al. 2017), in rabbits (Baiomy and Mansour 2016), and in fetal tissues (El-Aziz et al. 2018).

Other different types of herbal plants

Bharavi et al. (2011) reported the effect of using different types (holy basil, tulasi (*O. sanctum*) Satavar, shatavari, or shatamull (*Asparagus recemosus*), Kalmegh (*Andrographis paniculata*), Shilajith (*Asphaltum panjabinum*), Gurmar, Merasingi, small Indian ipecac (*Gymnema sylvestre*)) of herbal adaptogens against cadmium-induced oxidative stress and toxicity in broiler chicks. They concluded that using of such adaptogens reduced the bioaccumulation of cadmium in liver and kidney with subsequent decrease in oxidative damage.

Conclusion

Cd is a heavy metal that enters the body systems through different ways and then causes adverse influences on various tissues by prompting oxidative stress. Cd is one of the most toxic and carcinogenic heavy metals that can be mainly distributed to the liver, kidney, and testes, causing severe acute or chronic tissue injury. Cd accumulation in the body leads to organ dysfunction that manifests as renal proximal tubular cell injury, mild necrosis, inflammation, and hepatic endothelial cells. The exposure to higher levels of Cd has adverse effects on animal and poultry, including decreased growth rate and feed utilization. Many herbal plants, herbal extracts, and their active constituents can possess antioxidant, immunomodulation, and other beneficial health effects which can make them an effective strategy in preventing or reducing the bad impacts of Cd.

Recommendation

The herbs and herbal extracts presented in this review provide an alternative treatment opportunity in treating the damage caused by Cd poisoning. Further investigations would be required to demonstrate the action mechanism of these herbs in inhibiting Cd poisoning.

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